Harshith Kurakula

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SUMMARY

Software Engineer with 3+ years of experience in building automation tools, managing distributed systems, optimizing database performance, and improving application reliability in cloud environments. Proven expertise in streamlining workflows, enhancing system performance and scaling systems effectively. Skilled in Python, Java, SQL, AWS, CI/CD pipelines and Docker. Actively seeking summer internships in 2026.

EDUCATION

Arizona State University — *Tempe, AZ, USA*
Master of Science in Computer Science, **Current GPA - 4.00 / 4.00** — **August 2025 – Present**

Vellore Institute of Technology — *Vellore, TN, India*
Bachelor of Technology in Computer Science and Engineering, **GPA - 3.84 / 4.00** — **July 2018 – June 2022**

Relevant Coursework: Algorithms, Data Structures, Database Systems, Object-Oriented Programming, Web Mining, Machine Learning, Software Engineering, Web Development

SKILLS

Programming Languages:	Python, Java, SQL, C, C++, JavaScript, R
Cloud, DevOps & Databases:	AWS, Git, Docker, Kubernetes, Terraform, CI/CD, Splunk, MySQL, PostgreSQL, NoSQL
Web Development:	HTML5, CSS3, Flask, Django, REST APIs, Vue.js, Node.js
ML & Data Tools:	Scikit-learn, TensorFlow, NumPy, Pandas, OpenCV

EXPERIENCE

Chargebee Technologies — *Chennai, TN, India*

Senior Software Engineer (SRE Team) — **April 2025 – July 2025**

- Optimized **database connection management** by consolidating schema-level connection pools into a **shared pool per AWS RDS instance**, reducing connection spikes and idle connections by **20%** and improving **resource utilization, scalability, and system stability**.
- Automated **configuration onboarding** through a **centralized database configuration management system**, enabling consistent configuration across microservices and reducing **manual effort by 40% during infrastructure provisioning**.

Software Engineer (SRE Team) — **August 2022 – March 2025**

- Led **AWS RDS database upgrade initiatives** using **Blue-Green deployment strategies**, coordinating planning, execution, and validation to achieve **near-zero downtime**, preserve **data integrity**, and **reduce service disruption** by **90%** across critical production systems.
- Built an **automated schema comparison tool** using **Java** and **SQL** to detect database inconsistencies across environments, capturing **all schema mismatches** while **reducing manual debugging** time by **30%** and **improving overall production reliability**.
- Implemented an **API feature** for an internal database migration tool to **automate schema changes** following RDS provisioning and sharding activities. This streamlined setup **reduced manual intervention by 25%** and **improved system efficiency and reliability**.
- Collaborated with **cross-functional engineering, platform, and product teams** to embed **SRE best practices**, reducing **incident recurrence by 30%**, improving **system stability**, and enabling scalable **multi-region deployments**.

Software Engineer Intern (SRE Team) — **January 2022 – July 2022**

- Engineered a **centralized observability and monitoring dashboard** integrating **Splunk, ELK Stack, and database logs**, enabling proactive incident detection and reducing **MTTR by 25%** while maintaining **99.9% system availability**.
- Spearheaded **infrastructure provisioning automation** using AWS RDS, DynamoDB, and CloudFormation, reducing **manual configuration effort by 40%**, accelerating **deployment time by 30%**, and improving **cross-environment reliability and consistency**.

PROJECTS

Multimodal Image Retrieval System *(Python, Keras, Flask, Vue.js*

- Designed and implemented a **multimodal deep learning retrieval pipeline** leveraging **ResNet (CNN)** for visual feature extraction and **LSTM-based text embeddings**, achieving **80–85% retrieval accuracy** by effectively aligning **visual and semantic representations**.
- Built an **end-to-end image–text retrieval workflow** integrating **advanced text preprocessing** and **image feature analysis**, significantly outperforming **traditional baseline models** in **search relevance and precision**.

Facial Emotion Recognition *(Python, TensorFlow, Keras, OpenCV)*

- Engineered a **real-time facial emotion recognition system** using **OpenCV** and **Deep Convolutional Neural Networks (DCNNs)** to classify **7 facial expressions**, optimizing the model to capture **rapid facial dynamics** with higher accuracy than **traditional baselines**.
- Developed a **real-time alerting and data transmission module** to stream **emotion predictions and geolocation metadata**, enabling **immediate actionable insights** for **remote monitoring and safety applications**.